Appello 2 - soluzioni

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Analisi

1) a) $\dfrac{f(x,0)-f(0,0)}{x} = \dfrac{x^3/x^2 - 0}{x} = 1 \xrightarrow[x\to 0]{} 1$

$\dfrac{f(0,y)-f(0,0)}{y} = \dfrac{-3y^3/y^2 - 0}{y} = -3 \to -3$

$\Rightarrow \boxed{\nabla f(0,0) = (1,-3)}$

b) f diff in $(0,0) \Leftrightarrow$

$f(x,y) - f(0,0) = \nabla f(0,0)\cdot(x,y) + R(x,y)$

con $\lim_{(x,y)\to(0,0)} \dfrac{R(x,y)}{\sqrt{x^2+y^2}} = 0.$

È $f(x,y) - f(0,0) - \nabla f(0,0)(x,y) =$

$= \dfrac{x^3-3y^3}{x^2+y^2} - (x-3y) = \dfrac{x^3-3y^3-(x-3y)(x^2+y^2)}{x^2+y^2}$

$= \dfrac{\cancel{x^3}-\cancel{3y^3}-(\cancel{x^3}-3yx^2+xy^2-\cancel{3y^3})}{x^2+y^2}$

$= \dfrac{3yx^2-xy^2}{x^2+y^2} = \dfrac{xy}{x^2+y^2}(3x-y)$

ed è $\dfrac{\frac{xy}{x^2+y^2}(3x-y)}{\sqrt{x^2+y^2}} = \dfrac{xy(3x-y)}{(x^2+y^2)\sqrt{x^2+y^2}}$

Quindi f è differenziabile in $(0,0)$

$\Leftrightarrow \lim \dfrac{xy(3x-y)}{(x^2+y^2)\sqrt{x^2+y^2}} = 0.$

$(x,y)\to(0,0)$ $\quad (x^2+y^2)\sqrt{x^2+y^2}$ 1

Ma, se $y=x$ è $\dfrac{xy(3x-y)}{(x^2+y^2)\sqrt{x^2+y^2}} = \dfrac{2x^3}{2x^2\sqrt{2}|x|} \xrightarrow[x\to 0]{} 0$

$\Rightarrow$ $\boxed{f \text{ NON è differenziabile in } (0,0).}$

2. $yy' = 2x-1 \Rightarrow \dfrac{d}{dx}\left(\dfrac{1}{2}y^2\right) = 2x-1$
$= \dfrac{d}{dx}(x^2-x)$

$\Rightarrow \dfrac{1}{2}y^2(x) = x^2-x+c \qquad y^2(x) = 2x^2-2x+C$

$y(x) = \pm\sqrt{2x^2-2x+C}$.

È $y(0) = \pm\sqrt{C} = -1 \Leftrightarrow$ segno $-$ e

$C=1$:

$\boxed{y(x) = -\sqrt{2x^2-2x+1}}$, da cui

$\boxed{y(1) = -1.}$

3. $\displaystyle\int_{\{x^2+y^2\le 1} |x|y^2\,dx\,dy = 2\int_{\substack{\{x^2+y^2\le 1\\ x>0}} xy^2\,dx\,dy$

Poniamo $X=2x$, $Y=y$: $\begin{cases} x = \dfrac{X}{2} \\ y = Y \end{cases}$:

$\displaystyle 2\int_{\substack{X^2+Y^2\le 1\\ X>0}} \frac{X}{2}Y^2\,\frac{1}{2}\,dX\,dY = \frac{1}{2}\int_{\substack{X^2+Y^2\\ X>0}} XY^2\,dX\,dY$

$$= \left(\begin{array}{l} x = \rho \cos t \\ y = \rho \sin t, \quad t \in [0, \pi], \quad \rho \in [0,1] \end{array} \right)$$

π !

$$\frac{1}{2} \int_{\substack{t \in [0,\pi] \\ \rho \in [0,1]}} \rho^4 \cos t \sin^2 t \, d\rho \, dt$$

$$= \frac{1}{2} \left(\int_0^1 \rho^4 \, d\rho \right) \left[-\frac{1}{3} \sin^3 t \right]_0^{\pi} = \frac{1}{10} \cdot \frac{2}{3} = \boxed{\frac{1}{15}}$$

4. $\nabla f(x,y) = (2x + 3x^2 y, \; 2y + x^3) = (0,0)$

$$\Leftrightarrow \begin{cases} x(2 + 3xy) = 0 \\ 2y + x^3 = 0 \end{cases} \Rightarrow y = -x^3/2$$

$x(2 - 3x^4/2) = 0$

$\Rightarrow \left(x = 0 \right.$ o $x = \pm\sqrt[4]{\frac{4}{3}}$ e $y = -x^3/2$

Quindi i punti critici sono

- $x = 0$ e $y = 0$
- $x = \sqrt[4]{\frac{4}{3}}$, $y = -\frac{1}{2} \frac{4^{3/4}}{3^{3/4}}$
- $x = -\sqrt[4]{4/3}$ $\quad y = \frac{1}{2} \frac{4^{3/4}}{3^{3/4}}$

$$\text{Hess} f(x,y) = \begin{pmatrix} 2 + 6xy & 3x^2 \\ 3x^2 & 2 \end{pmatrix}$$

$\det \text{Hess} f(x,y) = 4 + 12xy - 6x^4$

Nei pti critici:

$\det \text{Hess} f(x,y) = 4 - 12x^4$

- $(0,0)$: $2>0$, det Hess $= 4>0$: $(0,0)$ <u>min loc. stretto</u>
- $\pm\left(\sqrt[4]{4/3}, -\frac{1}{2}\sqrt[4]{\frac{4^3}{3^3}}\right)$: $2+6xy = 2-3\cdot\sqrt[4]{\frac{4^4}{3^4}} = 2-3\cdot\frac{4}{3} < 0$

det Hess $= 4+12xy-6x^4 = 4-12x^4 = 4-12\frac{4}{3} = 4-16 = -12 < 0$

$\Rightarrow$ entrambi punti di <u>sella</u>.

Probabilità

1. (a) $X \sim B(800, 0.001)$

(b) $Y \sim P_o(800 \times 0.001) = P_o(0.8)$

$P(X=0) \approx P(Y=0) = e^{-0.8}$.

2. X_i = peso della cassa i

La probabilità cercata è

$P(X_1 + \dots + X_{49} < 9800)$

$X_1 + \dots + X_{49} \approx N(\mu, \sigma^2)$ (Th. c. Limite)

con $\mu = 49 \times 205 = 10.045$

$\sigma^2 = 49 \times 15^2$

$P(X_1 + \dots + X_{49} < 9800) \approx P(10.045 + 7.15Z < 9800)$

$= P(7.15Z < -245) = P\left(Z < -\frac{245}{7.15}\right) = \Phi\left(-\frac{35}{15}\right) = \Phi\left(-\frac{7}{3}\right)$

$= 1 - \Phi\left(\frac{7}{3}\right) = 1 - 0.9893 = 0.0107$.

3. (a) $f_X(x) = \begin{cases} 0 & x \le 0 \\ 6e^{-2x}\underbrace{\int_0^\infty e^{-3y}\,dy}_{} & x > 0 \end{cases} = \begin{cases} 0 & x \le 0 \\ 2e^{-2x} & x > 0 \end{cases}$

$$f_Y(y) = \begin{cases} 0 & \text{se } y \le 0 \\ 6e^{-3y} \underbrace{\int_0^\infty e^{-2x}\,dx}_{\frac{1}{2}} & y > 0 \end{cases} = \begin{cases} 0 & y \le 0 \\ 3e^{-3y} & y > 0 \end{cases}$$

X, Y <u>sono</u> indipendenti perché

$f_{X,Y}(x,y) = f_X(x) f_Y(y) \quad \forall x, y.$

(b) Si riconosce che $X \sim Exp(2)$,
$Y \sim Exp(3)$.

(c) $$P(X+Y<1) = \int_{\substack{0<x \\ 0<y \\ x+y<1}} 6e^{2x} e^{-3y}\,dx\,dy$$

$$= 6 \int_0^1 e^{-2x} \int_0^{1-x} e^{-3y}\,dy\,dx$$

$$= 6 \int_0^1 e^{-2x} \left[-\frac{1}{3} e^{-3y}\right]_{y=0}^{y=1-x} dx$$

$$= 2 \int_0^1 e^{-2x} \left(1 - e^{-3(1-x)}\right) dx$$

$$= 2 \int_0^1 e^{-2x} - e^{-3} e^{x}\,dx$$

$$= 2\left[-\frac{1}{2}e^{-2x} - e^{-3}e^{x}\right]_0^1$$

$$= 2\left(-\frac{1}{2}e^{-2} - e^{-2} + \frac{1}{2} + e^{-3}\right) = -3e^{-2} + 1 + 2e^{-3} \approx 0.69$$